

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO5: Apply computer vision techniques to real-world applications such as face recognition, tracking, medical imaging, and autonomous systems. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 20 Total Marks:20 Annexure A

Mock Interview

Code of Conduct:

I Shri Harshini Devi A - 192524104 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge	
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem solving ability; requires guidance	Unable to solve or apply concepts	
Analytical Thinking	15	Analyzes questions critically and provides well structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated	
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively	

Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism	
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately	

Annexure A

Mock Interview

1. What is Eigenfaces?
2. Define PCA in image processing.
3. What is contour-based shape representation?
4. What is region-based shape representation?
5. How does an active contour (snake) work?
6. How does a level set method work?
7. What is the Hough Transform?
8. How are Fourier descriptors used in shape analysis?
9. How are wavelet descriptors used in shape analysis?
10. What is a particle filter?
11. How is foreground–background separation done in surveillance systems?
12. How do multi-camera systems track pedestrians?
13. What is a Convolutional Neural Network (CNN)?
14. How does a Vision Transformer (ViT) work?
15. What is a Generative Adversarial Network (GAN)?
16. How is a Support Vector Machine (SVM) used for image classification?
17. How are Hidden Markov Models (HMM) applied in activity recognition?
18. What is a Gaussian Mixture Model (GMM)?
19. How is graph-based learning applied to 3D shape

correspondence?

20. How is road and lane detection performed in autonomous driving?

1. What is Eigenfaces?

Eigenfaces is a face recognition technique based on **PCA (Principal Component Analysis)**. It represents faces using a small number of important features called eigenfaces and compares a new face with them.

2. Define PCA in image processing.

PCA is a dimensionality-reduction technique that transforms high-dimensional image data into fewer important features while preserving maximum variance. It is commonly used for **feature extraction and face recognition**.

3. What is contour-based shape representation?

Contour-based representation describes an object using its **boundary or outline**. Only the external shape is considered, not the internal pixels.

Example: Representing a leaf using its boundary points.

4. What is region-based shape representation?

Region-based representation describes an object using the **entire region inside its boundary**. It considers properties such as area, texture, and pixel distribution.

5. How does an active contour (snake) work?

An **active contour**, or snake, is a flexible curve that moves toward an object's boundary. It uses **internal forces** to keep the curve smooth and **external image forces** to attract it toward edges.

6. How does a level set method work?

A **level set method** represents a contour as a mathematical function.

The contour evolves automatically according to image information until it reaches the desired object boundary. It can naturally handle **splitting and merging of objects**.

7. What is the Hough Transform?

The **Hough Transform** is a technique used to detect geometric shapes such as **lines, circles, and curves** in images. It converts image points into a parameter space and finds shapes by detecting peaks.

8. How are Fourier descriptors used in shape analysis?

Fourier descriptors represent an object's **boundary using Fourier coefficients**. They convert boundary information into frequency components, which can be used to **compare and classify shapes**.

9. How are wavelet descriptors used in shape analysis?

Wavelet descriptors represent shapes at **different scales and resolutions**. They are useful for capturing both large-scale and fine details of an object's boundary.

10. What is a particle filter?

A **particle filter** is a tracking algorithm that represents the possible state of an object using many random samples called **particles**. It is useful for tracking objects under **noise, uncertainty, and occlusion**.

11. How is foreground–background separation done in surveillance systems?

It separates moving objects from the static background using techniques such as **background subtraction, frame differencing, or Gaussian Mixture Models (GMM)**. The detected foreground is then used to identify moving objects.

12. How do multi-camera systems track pedestrians?

Multiple cameras observe the same area from different viewpoints. The system **detects pedestrians in each camera, matches their features/identities across cameras, and combines their positions**

to maintain continuous tracking.

13. What is a Convolutional Neural Network (CNN)?

A **CNN** is a deep learning model mainly used for image processing. It uses **convolution layers to extract features**, pooling layers to reduce dimensions, and fully connected layers for classification.

14. How does a Vision Transformer (ViT) work?

A **Vision Transformer** divides an image into small **patches**, converts them into embeddings, and processes them using a **Transformer with self-attention**. It learns relationships between different parts of the image.

15. What is a Generative Adversarial Network (GAN)?

A **GAN** consists of two neural networks: a **Generator** and a **Discriminator**. The generator creates fake data, while the discriminator tries to distinguish real data from fake data. Both improve through competition.

16. How is an SVM used for image classification?

Support Vector Machine (SVM) classifies images by finding an optimal **decision boundary (hyperplane)** between different classes. Image features are given as input to the SVM for classification.

17. How are Hidden Markov Models (HMM) applied in activity recognition?

An **HMM** models activities as a sequence of hidden states. Observations such as **human movements or poses** are used to determine the most likely activity, such as walking, running, or sitting.

18. What is a Gaussian Mixture Model (GMM)?

A **GMM** represents data as a mixture of multiple **Gaussian probability distributions**. In computer vision, it is commonly used for **background modeling and foreground detection**.

19. How is graph-based learning applied to 3D shape correspondence?

3D shapes are represented as **graphs**, where vertices represent points or regions and edges represent relationships between them. Graph-based learning finds corresponding points or regions between two 3D shapes.

20. How is road and lane detection performed in autonomous driving?

A typical pipeline is:

Video frame → Grayscale → Gaussian smoothing → Canny edge detection → Region of Interest → Hough Transform → Lane lines

The detected lane boundaries are then highlighted on the image. For curved or complex roads, **deep learning-based segmentation and object detection** can also be used.